

CHAPTER 8

Cancer, Heart Attacks, and a Shorter Life

Sleep Deprivation and the Body

I was once fond of saying, “Sleep is the third pillar of good health, alongside diet and exercise.” I have changed my tune. Sleep is more than a pillar; it is the foundation on which the other two health bastions sit. Take away the bedrock of sleep, or weaken it just a little, and careful eating or physical exercise become less than effective, as we shall see.

Yet the insidious impact of sleep loss on health runs much deeper. Every major system, tissue, and organ of your body suffers when sleep becomes short. No aspect of your health can retreat at the sign of sleep loss and escape unharmed. Like water from a burst pipe in your home, the effects of sleep deprivation will seep into every nook and cranny of biology, down into your cells, even altering your most fundamental self—your DNA.

Widening the lens of focus, there are more than twenty large-scale epidemiological studies that have tracked millions of people over many decades, all of which report the same clear relationship: the shorter your sleep, the shorter your life. The leading causes of disease and death in developed nations—diseases that are crippling health-care systems, such as heart disease, obesity, dementia, diabetes, and cancer—all have recognized causal links to a lack of sleep.

This chapter describes, uncomfortably, the many and varied ways in which insufficient sleep proves ruinous to all the major physiological systems of the human body: cardiovascular, metabolic, immune, reproductive.

SLEEP LOSS AND THE CARDIOVASCULAR SYSTEM